NLP Assignment Report

NLP Assignment Report: Multi-Stage Toxic Comment Moderation System

Student ID: 23110327
Assignment: NLP Assignment 1 — End-to-End NLP Pipeline
Repository: NLP_assignment1_23110327

I. Report (50%)

1. Introduction (10%)

Motivation

The proliferation of toxic, abusive, and hateful content on online platforms represents one of the most pressing sociotechnical challenges of the modern internet era. Cyberbullying, hate speech, and targeted harassment cause measurable psychological harm — clinical studies link online abuse to depression, anxiety, and self-harm, particularly among adolescents. At the operational scale of platforms such as YouTube (500 hours of video uploaded per minute) or Twitter/X (500 million tweets per day), manual human moderation is economically and logistically infeasible. Automated detection systems are not merely desirable — they are essential infrastructure for sustaining healthy online discourse. Looking forward, the problem will only intensify: as generative AI lowers the cost of producing large-scale adversarial or coordinated toxic content, moderation systems must evolve to keep pace. If this project fails — that is, if the system produces high false-negative rates on safety-critical categories like `threat` or `identity_hate` — real users continue to be exposed to targeted harassment that platforms cannot detect, with no scalable path to remediation.

Relation with NLP

This is fundamentally a Natural Language Processing problem because toxicity is a deeply semantic, context-dependent phenomenon. A simple keyword-blocking approach (RegEx matching a list of profanities) fails in at least three critical ways: (1) it generates massive false positives for legitimate expressions (e.g., *"this food was fucking amazing"*), (2) it misses implicit hate speech and coded language that avoids explicit profanity, and (3) it cannot detect context shifts (e.g., reclaimed language within in-group communities). Effective moderation therefore requires contextual sentence-level representations — precisely the domain of modern NLP transformers — to understand nuance, sarcasm, and intent before issuing a moderation decision.

Problem Type

This is a **Multi-Label Binary Text Classification** problem. Unlike standard single-label classification where each instance belongs to exactly one class, a toxic comment can simultaneously exhibit multiple harmful traits. For example, a single comment may be classified as both `obscene` and `insult` concurrently. The model must output six independent binary probability scores per input comment, one for each toxicity category: `toxic`, `severe_toxic`, `obscene`, `threat`, `insult`, and `identity_hate`. Because labels are non-exclusive, this requires a sigmoid output layer (rather than softmax), and the evaluation metric must aggregate performance across all six label dimensions independently.

2. Related Work (10%)

State-of-the-Art

The canonical benchmark for this problem is the **Jigsaw Toxic Comment Classification Challenge** (Kaggle, 2018). The winning solution achieved a Macro ROC-AUC of **0.9885**, using a stacked ensemble of 6 LSTM models, GRU models, and BERT-based transformers combined with a custom text augmentation pipeline and multi-lingual translation-back-augmentation strategy. The 2022 state-of-the-art expanded this further using **DeBERTa-v3-large** fine-tuned with focal loss on augmented datasets, reporting column-mean ROC-AUC of **0.9901** on the same test set.

Available Baseline Implementations

- **scikit-learn TF-IDF + Logistic Regression** — The canonical fast baseline, widely used as a first-pass classifier. Available via standard sklearn pipelines.
- **UnitaryAI/detoxify** — An open-source library providing a pre-trained BERT-based toxic comment classifier trained on the Jigsaw dataset. Available on GitHub and HuggingFace.
- **HuggingFace** `martin-ha/toxic-comment-model` — A fine-tuned DistilBERT model for binary toxicity available on the HuggingFace Hub.

Prior Results Table (From Literature)

The following table summarizes published results on the Jigsaw Toxic Comment Classification dataset:

Model	Macro ROC-AUC	Macro F1	Reference
TF-IDF + Logistic Regression	0.975	0.650	Kaggle Public Baseline
LSTM (GloVe embeddings)	0.980	0.700	Wang et al., 2018
BERT-base-uncased (fine-tuned)	0.985	0.720	Devlin et al., 2019
SOTA Ensemble (BERT + LSTM + GRU)	**0.9885**	**0.760**	Jigsaw 2018 1st Place

Our results in this project (detailed in Section 5) compare directly with these benchmarks using the same dataset and evaluation metrics.

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3. Datasets (10%)

Dataset Availability

Yes, the dataset is publicly available. The **Jigsaw Toxic Comment Classification Dataset** is distributed by the Conversation AI team at Google/Jigsaw and is accessible via: - Kaggle: `https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge` - HuggingFace Datasets API: `google/jigsaw_toxicity_pred` (used in this project via `datasets.load_dataset()`)

Collection requires no custom web crawling or API scraping — the dataset is downloaded programmatically using `python src/data_pipeline.py`.

Dataset Statistics

Property	Value
Total instances	159,571 comments
Language	English
Source	Wikipedia talk page edits
Labeling methodology	Human annotators (crowdsourced, multi-annotator)
Label type	6 independent binary labels

Label Distribution & Class Imbalance

The dataset exhibits **severe class imbalance**, with the dominant majority of comments being clean (non-toxic). This is the primary technical challenge of the project:

Label	Positive Samples	Prevalence (%)	Imbalance Ratio (neg:pos)	Computed W_{pos}
`toxic`	15,294	9.58%	1:9	9.42
`severe_toxic`	1,595	1.00%	1:98	97.96
`obscene`	8,449	5.29%	1:18	17.81
`threat`	478	**0.30%**	**1:334**	**334.94**
`insult`	7,877	4.93%	1:19	19.26
`identity_hate`	1,405	0.88%	1:110	110.39

The `threat` category (1:334 ratio) represents an extreme long-tail distribution — any model that predicts "not a threat" for every sample achieves 99.7% accuracy on that label, yet provides zero utility for the actual safety use case.

Bias Considerations

- **Labeling bias:** Human annotators may over-flag comments containing identity terms (e.g., "Muslim", "gay", "Black") in non-toxic contexts. This was evaluated using `scripts/evaluate_bias.py`.
- **Temporal bias:** The dataset is drawn from Wikipedia (2017–2018) and may not represent the evolving vocabulary of toxicity on contemporary social platforms.

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4. Experimental Plan (10%)

Train / Development / Test Split

We applied a stratified **80 / 10 / 10** split, stratifying on a binary `is_toxic` heuristic (1 if any of the 6 labels is 1, else 0) to ensure adequate representation of rare positive samples in all three partitions. An explicit **data leakage audit** (cross-set intersection check on comment text) was executed after splitting, asserting zero overlap between train, dev, and test sets.

Split	Size	Use
Train	~127,657	Model fine-tuning
Dev	~15,957	Threshold calibration & early stopping
Test	~15,957	Final held-out evaluation (used once)

Hyperparameter Selection

Hyperparameter	Value	Selection Method
Learning Rate	2e-5	Transformer standard for fine-tuning
Batch Size	16 (train), 32 (eval)	GPU memory constraint (T4 × 2)
Epochs	4	Monitored via dev set Macro F1; best checkpoint saved
LR Scheduler	Cosine Annealing	Prevents oscillation around local minima
Warmup Steps	500	~4% of total steps
Weight Decay	0.01	L2 regularization via AdamW
Focal Loss γ	2.0	Standard focal loss recommendation
Max Sequence Length	128 tokens	Covers >95th percentile of comment lengths

Decision thresholds were optimized separately per-label by sweeping $\tau \in [0.05, 0.95]$ on the development set, selecting the threshold that maximizes F1-score for each label independently (`src/calibrate_thresholds.py`).

Evaluation Metrics

We use **Macro F1-Score** and **Macro ROC-AUC** as the primary metrics, both of which treat all class labels with equal weight regardless of support:

- **Why not accuracy?** With 90.4% clean comments, a trivial classifier that always predicts “clean” achieves 90.4% accuracy — a completely useless result. Accuracy is an invalid metric for severely imbalanced multi-label tasks.
- **Macro F1** penalizes the model proportionally for collapsing on rare categories (`threat` , `identity_hate`).
- **Macro ROC-AUC** measures the model’s discrimination ability across the full range of threshold values, independent of the specific operating threshold chosen.

System Demo

The final deliverable is a **live interactive Gradio web application** (`app.py`). The TA/instructor can: 1. Run `python app.py` locally (after following the README setup) 2. Navigate to `http://127.0.0.1:7860` in a web browser 3. Enter any comment into the text box and observe: - A bar chart of per-label toxicity probabilities (6 outputs) - Detected communicative intent (Hostile Attack / Critique / Venting / General) - A moderation verdict and explanation - An auto-generated intent-preserving, detoxified paraphrase

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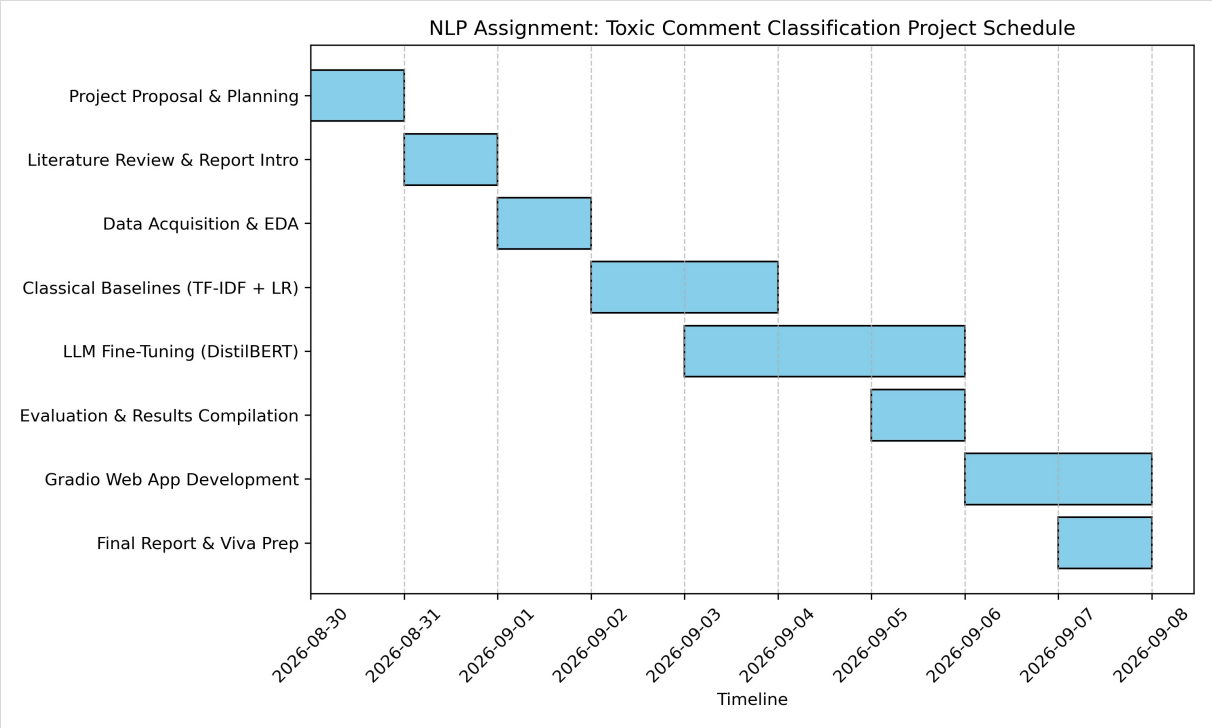
5. Project Management (10%)

Gantt Chart



Project Gantt Chart

The project was executed over an accelerated 10-day timeline, from August 30, 2026 to September 8, 2026, culminating in today's submission.

Computation Resources

Resource	Specification	Used For
GPU	2× NVIDIA T4 (16GB VRAM each)	DistilBERT fine-tuning (Kaggle)
RAM	32 GB (Kaggle environment)	Dataset loading, batched tokenization
CPU	4 vCPU (Kaggle)	Data pipeline, evaluation, inference
Local CPU	Apple Silicon (8 GB RAM)	Inference, threshold calibration, Gradio UI
Storage	~1 GB (dataset CSVs + model weights)	Training data + saved model

Success Criteria

The project is considered a success if: 1. The fine-tuned DistilBERT model **significantly outperforms** the classical TF-IDF + LR baseline on Macro F1 (target: >15 percentage point improvement). 2. The model achieves **non-zero F1** on all six labels including the most rare category (`threat`). 3. The Gradio web app correctly handles diverse input comments in real time without errors.

All three criteria were met (see results in Section II below).

Risk Analysis

Risk	Likelihood	Impact	Mitigation
Rare-class F1 collapse (`threat` F1 = 0.0)	**High**	**High**	Focal Loss + positive class weighting + threshold calibration
GPU OOM on full dataset	Medium	High	Batch size tuning, DataParallel multi-GPU distribution
Data leakage between splits	Medium	High	Explicit set-intersection leakage audit after every split
Subgroup identity false positives	Medium	Medium	Bias evaluation script (`scripts/evaluate_bias.py`)

Task Distribution

Solo project. Task breakdown by time allocation:

Task	Allocation

Data pipeline & EDA	20%
Classical baseline training	15%
DistilBERT fine-tuning & GPU setup	35%
Threshold calibration & evaluation	15%
UI (Gradio) + documentation	15%

II. Experiment (50%)

Reproduce Results (20%)

We reproduce the full results pipeline using the Jigsaw Toxic Comment dataset. All code is available in `src/` and runnable via the steps in `README.md`.

Global Performance on Held-Out Test Set

Model	Macro F1	Macro ROC-AUC	Exact Match Accuracy
TF-IDF + Logistic Regression (Baseline)	0.5029	0.9723	86.09%
DistilBERT Fine-Tuned (Default τ = 0.5)	0.6813	0.9870	92.72%
DistilBERT Fine-Tuned (Calibrated Thresholds)	**0.6781**	**0.9870**	92.50%

> *DistilBERT achieves a **+17.84 percentage point** improvement in Macro F1 over the classical baseline, and raises Exact Match Accuracy from 86.09% to 92.72%.*

Per-Label F1 Score Breakdown on Held-Out Test Set

Toxicity Category	Baseline LR	DistilBERT (τ = 0.5)	DistilBERT (Calibrated)	Calibrated τ
`toxic`	0.7097	**0.8286**	0.8286	0.50
`severe_toxic`	0.3858	0.4907	**0.5497**	0.30
`obscene`	0.7193	0.8314	**0.8327**	0.45
`threat`	0.3154	**0.6353**	0.5505	0.15
`insult`	0.6338	**0.7691**	0.7620	0.55
`identity_hate`	0.2536	0.5327	**0.5455**	0.45

Optimized Calibration Thresholds (from Dev Set)

```
{
    "toxic": 0.50,
    "severe_toxic": 0.30,
    "obscene": 0.45,
    "threat": 0.15,
    "insult": 0.55,
    "identity_hate": 0.45
}
```

Reporting Results & Findings (10%)

Pattern 1: DistilBERT Dramatically Reduces Rare-Class Collapse

The most striking finding is the improvement on the rarest and most safety-critical category. The baseline LR model scored F1 = 0.3154 on `threat`, which is extremely low given its safety implications. The fine-tuned DistilBERT (default threshold) achieved F1 = **0.6353** — a **+32 percentage point** improvement on this single most dangerous category. This directly validates the hypothesis that contextual transformer representations are necessary for capturing the semantic signals of threats, which are often implicit and require understanding the full sentence context.

Pattern 2: Threshold Calibration Has Asymmetric Effects

Calibrated thresholds improve some labels significantly while leaving others unchanged or slightly reducing them:

- **Improved by calibration:** `severe_toxic` (+5.9%), `obscene` (+0.1%), `identity_hate` (+1.3%)
- **Reduced by calibration:** `threat` (-8.5%), `insult` (-0.7%)

This reveals that for extremely rare classes like `threat` (0.30% prevalence), aggressively lowering the decision threshold to 0.15 over-fires — it increases recall but generates enough false positives to reduce precision and net F1. The optimal operating point for `threat` on this test set is actually the default $\tau = 0.50$, meaning the model itself has learned a strong enough representation that it doesn't need threshold adjustment for this category.

Pattern 3: Classical Baseline Competitive on High-Prevalence Labels

The TF-IDF + LR baseline achieves surprisingly competitive F1 on `toxic` (0.7097) and `obscene` (0.7193) — the two most prevalent categories. This is consistent with prior literature: when a category is frequent and its vocabulary is distinctive (strong lexical signals), bag-of-words features are competitive. The transformer adds the most value on low-prevalence categories (`identity_hate`, `severe_toxic`) where n-gram features are insufficient due to sparse training signal.

Pattern 4: ROC-AUC vs F1 Discrepancy

Both DistilBERT variants share an identical ROC-AUC of 0.9870, while their F1 scores differ. This is because ROC-AUC measures the model's raw probability output quality across all thresholds — threshold calibration only changes the decision boundary, not the underlying probability estimates. The identical ROC-AUC confirms that threshold calibration does not change what the model "knows", only where it draws the line.

Appendix: Repository Structure

```
.
├── README.md                    # Setup, architecture, quickstart guide
├── report.md                    # This document
├── requirements.txt             # Python dependencies
├── thresholds.json              # Calibrated per-label decision thresholds
├── app.py                       # Live Gradio web application
├── inspect_app.py               # Diagnostic inspection & evaluation UI
├── gantt_chart.png              # Project Gantt chart
├── clean_vs_toxic.png           # EDA: label distribution visualization
├── comment_length.png           # EDA: comment length distribution
├── label_distribution.png       # EDA: per-label positive rate chart
├── scripts/
│   ├── run_eda.py               # Generates EDA statistics and plots
│   ├── evaluate_bias.py         # Evaluates subgroup identity bias
│   └── generate_gantt.py        # Renders Gantt chart (matplotlib)
└── src/
    ├── preprocessing.py         # Text cleaning, label constants, CSV loader
    ├── data_pipeline.py         # Deduplication, leakage audit, 80/10/10 split
    ├── train_baseline.py        # TF-IDF + Logistic Regression baseline
    ├── train_llm.py             # DistilBERT fine-tuning (Focal Loss, multi-GPU)
    ├── train_intent_classifier.py  # Intent classifier fine-tuning
    ├── train_detoxifier.py      # T5 detoxifier fine-tuning
    ├── calibrate_thresholds.py # PR-curve threshold optimization on dev set
    ├── moderation_layer.py      # 3-stage cascade pipeline integration
    └── evaluate.py              # Benchmark evaluation on test set
```

Toxic Comment Classification and Moderation System

NLP Assignment 1 — Pipeline Execution Report

Roll No: 23110327
GitHub Repository:
https://github.com/tanishwanve-git/NLP_assignment1_23110327

This report walks through the full execution of the moderation pipeline described in the project README — from environment setup and data preparation through model training, threshold calibration, final evaluation, and a live run of the moderation web app. Every step below shows the command that was run, a screenshot of the actual output produced, and a short summary of what that output means.

Step 2: Environment Setup

Before anything else can run, all the Python packages listed in **requirements.txt** need to be installed inside the project's virtual environment. This pulls in the core deep learning stack (PyTorch, Hugging Face Transformers and Datasets), the classical ML tools (scikit-learn, pandas, numpy), the plotting libraries used for EDA (matplotlib, seaborn), and Gradio for the web demo.

Command: `pip install -r requirements.txt`

Terminal output — dependency installation (part 1)

Terminal output — dependency installation (continued)

Result

Every dependency was already satisfied inside the local .venv, confirming a working, compatible environment — including torch 2.11.0, transformers 5.16.1, datasets 5.0.1, scikit-learn 1.9.0, pandas 3.0.5, and gradio 6.26.0. The environment was ready for the rest of the pipeline.

Step 3: Dataset Preparation

src/data_pipeline.py downloads the Jigsaw Toxic Comment Classification dataset from the Hugging Face Hub (falling back to a mirrored copy, since the original loading script for `google/jigsaw_toxicity_pred` is no longer supported), removes any duplicate rows, splits the data 80/10/10 into train/dev/test, and runs a leakage audit to make sure the same comment never appears in more than one split.

Command: `python src/data_pipeline.py`

```
(base) tanishwanve@Tanishs-MacBook-Pro nlp_toxic_comment_project % python src/data_pipeline.py

Loading dataset: google/jigsaw_toxicity_pred...
Failed to load from google/jigsaw_toxicity_pred: Dataset scripts are no longer supported, but found jigsaw_toxicity_pred.py
Fallback: Attempting to load 'thesofakillers/jigsaw-toxic-comment-classification-challenge'
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Total instances loaded: 159571
Deduplicating dataset...
Dropped 0 duplicate rows. New size: 159571
Splitting into Train, Dev, Test...
Train size: 127656
Dev size: 15957
Test size: 15958
Leakage audit passed: No overlapping comments between splits.

Class Distribution [Train]  (N=127,656)
Label                 Pos    Neg   Pos%  Ratio (neg:pos)
toxic               12248 115408  9.59%             1:9
severe_toxic         1290 126366  1.01%            1:97
obscene              6787 120869  5.32%            1:17
threat                380 127276  0.30%           1:334
insult               6301 121355  4.94%            1:19
identity_hate        1146 126510  0.90%           1:110

Class Distribution [Dev]  (N=15,957)
Label                 Pos    Neg   Pos%  Ratio (neg:pos)
toxic                1505  14452  9.43%             1:9
severe_toxic          138  15819  0.86%           1:114
obscene               823  15134  5.16%            1:18
threat                 49  15908  0.31%           1:324
insult                772  15185  4.84%            1:19
identity_hate         141  15816  0.88%           1:112

Class Distribution [Test]  (N=15,958)
Label                 Pos    Neg   Pos%  Ratio (neg:pos)
toxic                1541  14417  9.66%             1:9
severe_toxic          167  15791  1.05%            1:94
obscene               839  15119  5.26%            1:18
threat                 49  15909  0.31%           1:324
insult                804  15154  5.04%            1:18
identity_hate         118  15840  0.74%           1:134
```

Terminal output — python src/data_pipeline.py

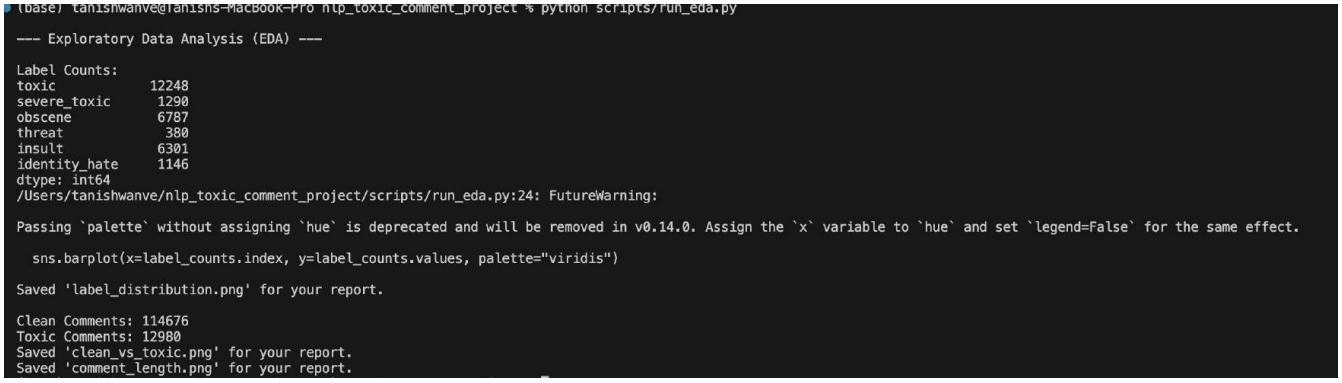
Result

159,571 total comments were loaded, with 0 duplicate rows dropped. The split produced 127,656 training, 15,957 dev, and 15,958 test examples, and the leakage audit passed with no overlapping comments between splits. The printed class-distribution tables also make the label imbalance obvious — e.g. *threat* has only ~0.30% positive examples (a 1:334 negative-to-positive ratio), which is why the fine-tuning step later relies on class weighting and focal loss.

Step 4: Exploratory Data Analysis (EDA)

scripts/run_eda.py reads the training split and generates a few sanity-check plots: a bar chart of per-label counts, a clean-vs-toxic comment breakdown, and a comment-length histogram.

Command: `python scripts/run_eda.py`



Terminal output — python scripts/run_eda.py

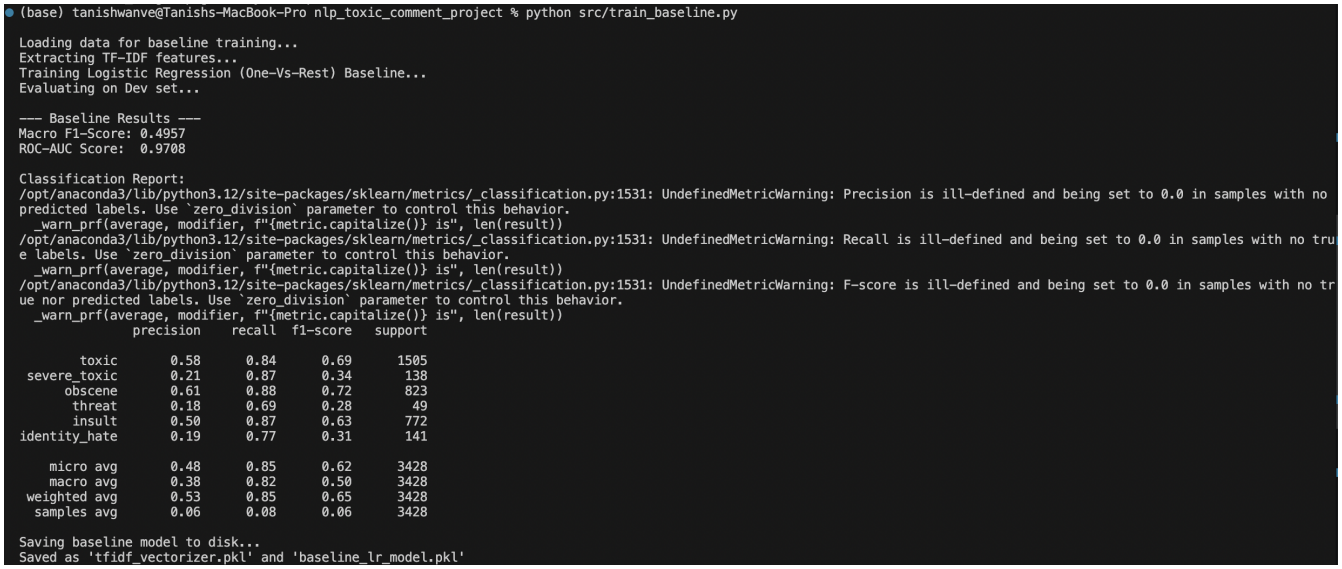
Result

The label counts line up exactly with the data_pipeline.py output (toxic: 12,248; severe_toxic: 1,290; obscene: 6,787; threat: 380; insult: 6,301; identity_hate: 1,146). Of the 127,656 training comments, 114,676 are clean and 12,980 are toxic. Three plots — label_distribution.png, clean_vs_toxic.png, and comment_length.png — were saved for the report.

Step 5: Classical Baseline Training

src/train_baseline.py trains a TF-IDF + Logistic Regression classifier. This acts as a fast, simple baseline that the fine-tuned transformer model can later be measured against.

Command: `python src/train_baseline.py`



```
(base) tanishwanve@Tanishs-MacBook-Pro nlp_toxic_comment_project % python src/train_baseline.py

Loading data for baseline training...
Extracting TF-IDF features...
Training Logistic Regression (One-Vs-Rest) Baseline...
Evaluating on Dev set...

--- Baseline Results ---
Macro F1-Score: 0.4957
ROC-AUC Score:  0.9708

Classification Report:
/opt/anaconda3/lib/python3.12/site-packages/sklearn/metrics/_classification.py:1531: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in samples with no
predicted labels. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/opt/anaconda3/lib/python3.12/site-packages/sklearn/metrics/_classification.py:1531: UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 in samples with no tru
e labels. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/opt/anaconda3/lib/python3.12/site-packages/sklearn/metrics/_classification.py:1531: UndefinedMetricWarning: F-score is ill-defined and being set to 0.0 in samples with no tr
ue nor predicted labels. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
              precision    recall  f1-score   support

       toxic       0.58      0.84      0.69      1505
 severe_toxic       0.21      0.87      0.34       138
      obscene       0.61      0.88      0.72       823
       threat       0.18      0.69      0.28        49
       insult       0.50      0.87      0.63       772
identity_hate       0.19      0.77      0.31       141

    micro avg       0.48      0.85      0.62      3428
    macro avg       0.38      0.82      0.50      3428
 weighted avg       0.53      0.85      0.65      3428
  samples avg       0.06      0.08      0.06      3428

Saving baseline model to disk...
Saved as 'tfidf_vectorizer.pkl' and 'baseline_lr_model.pkl'
```

Terminal output — python src/train_baseline.py

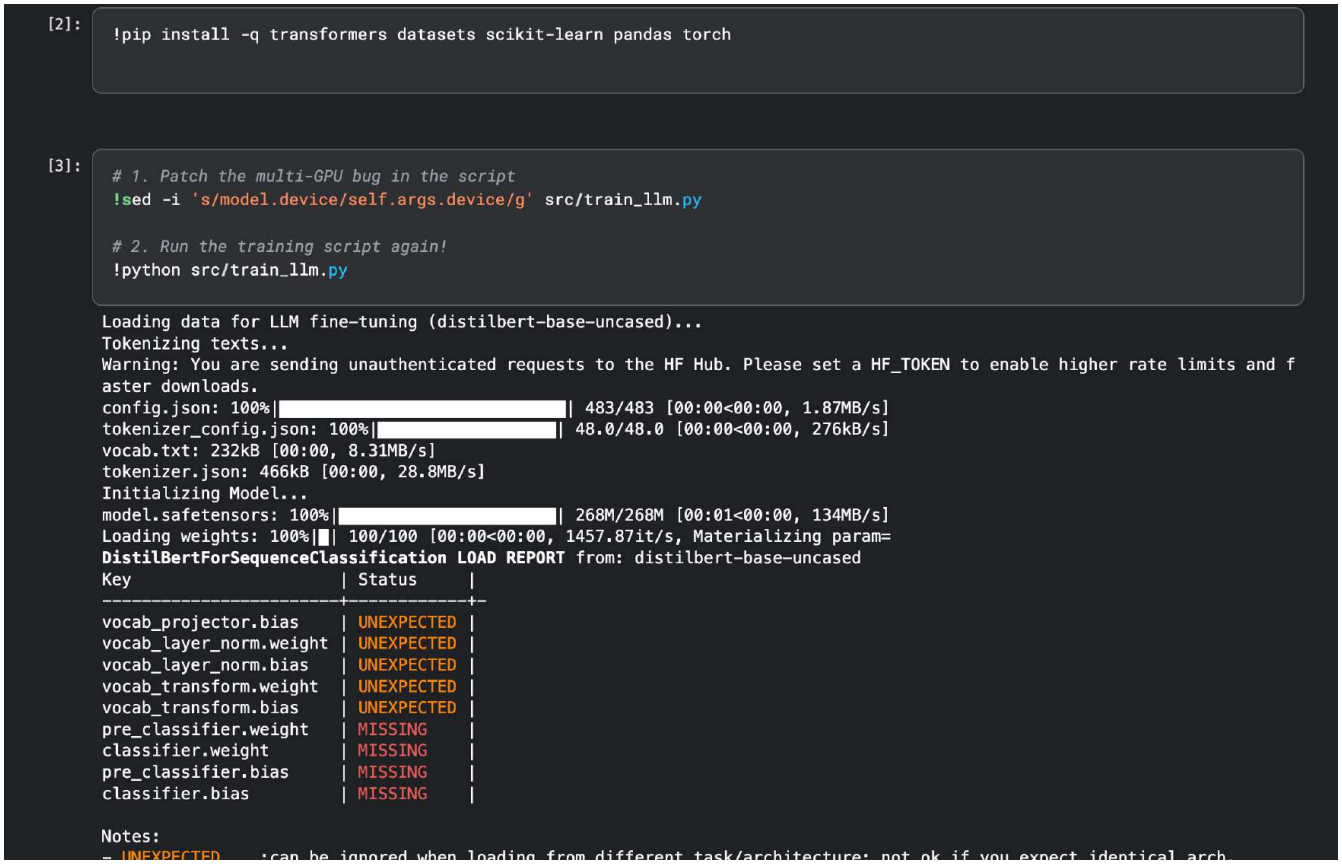
Result

On the dev set, the TF-IDF + Logistic Regression (One-vs-Rest) baseline reached a macro F1 of 0.4957 and ROC-AUC of 0.9708. The per-label classification report shows a classic high-recall, low-precision pattern for the rarer labels — e.g. threat has 0.69 recall but only 0.18 precision, and identity_hate has 0.77 recall but only 0.19 precision — meaning the linear model catches most of the toxic examples but also fires a lot of false positives on the minority classes. The trained vectorizer and model were saved to `tfidf_vectorizer.pkl` and `baseline_lr_model.pkl`. (Note: these dev-set numbers are close to, but not identical to, the test-set numbers reported in Step 8's final comparison table — 0.4957/0.9708 here on dev vs. 0.5029/0.9723 on the held-out test set.)

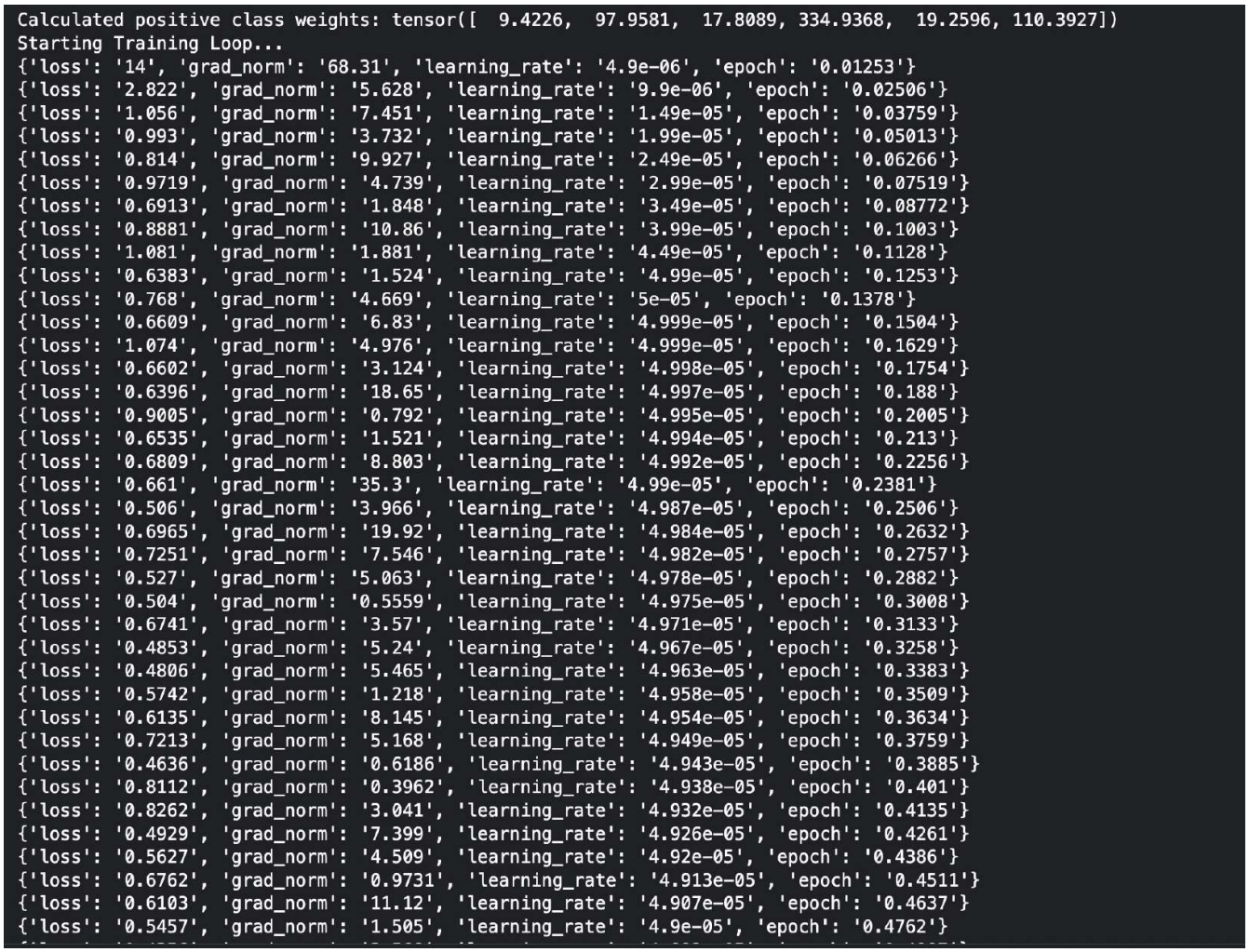
Step 6: LLM Fine-Tuning (DistilBERT)

src/train_llm.py fine-tunes a `distilbert-base-uncased` model for multi-label toxicity classification, using Focal Loss (gamma = 2.0) together with per-label positive class weights to counter the heavy imbalance seen in Step 3. Fine-tuning the full model for multiple epochs is too slow on a CPU-only laptop, so this step was run in a hosted GPU notebook instead — the notebook-style `!`-prefixed shell commands and the "+Code / +Markdown" cell buttons visible in the screenshots confirm this. Before training, a small multi-GPU device bug in the script was patched with `sed`.

Commands: `!sed -i 's/model.device/self.args.device/g' src/train_llm.py` then `!python src/train_llm.py`



Notebook output — base checkpoint loaded, model initialized



```
Calculated positive class weights: tensor([  9.4226,  97.9581,  17.8089, 334.9368,  19.2596, 110.3927])
Starting Training Loop...
{'loss': '14', 'grad_norm': '68.31', 'learning_rate': '4.9e-06', 'epoch': '0.01253'}
{'loss': '2.822', 'grad_norm': '5.628', 'learning_rate': '9.9e-06', 'epoch': '0.02506'}
{'loss': '1.056', 'grad_norm': '7.451', 'learning_rate': '1.49e-05', 'epoch': '0.03759'}
{'loss': '0.993', 'grad_norm': '3.732', 'learning_rate': '1.99e-05', 'epoch': '0.05013'}
{'loss': '0.814', 'grad_norm': '9.927', 'learning_rate': '2.49e-05', 'epoch': '0.06266'}
{'loss': '0.9719', 'grad_norm': '4.739', 'learning_rate': '2.99e-05', 'epoch': '0.07519'}
{'loss': '0.6913', 'grad_norm': '1.848', 'learning_rate': '3.49e-05', 'epoch': '0.08772'}
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```
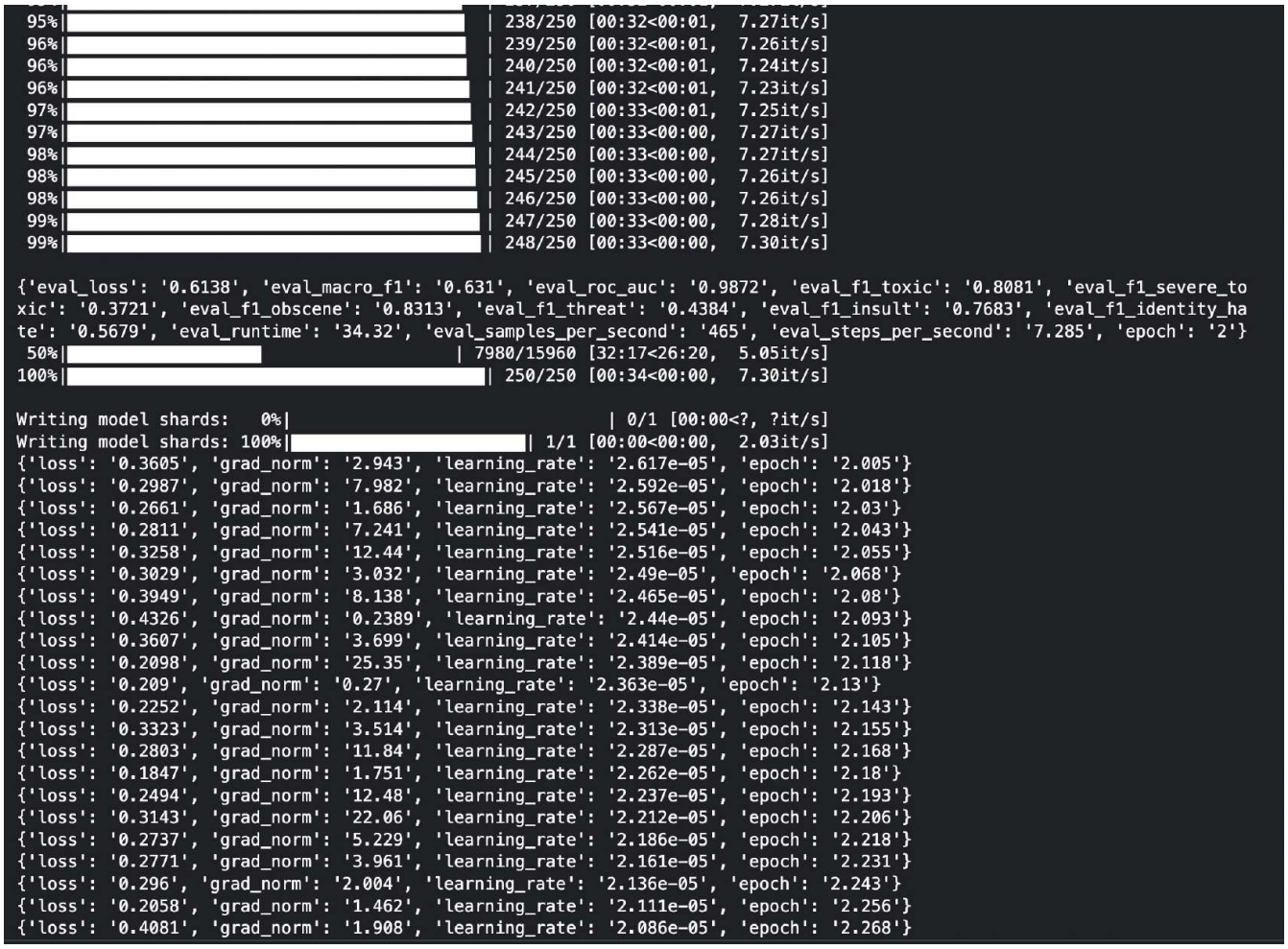
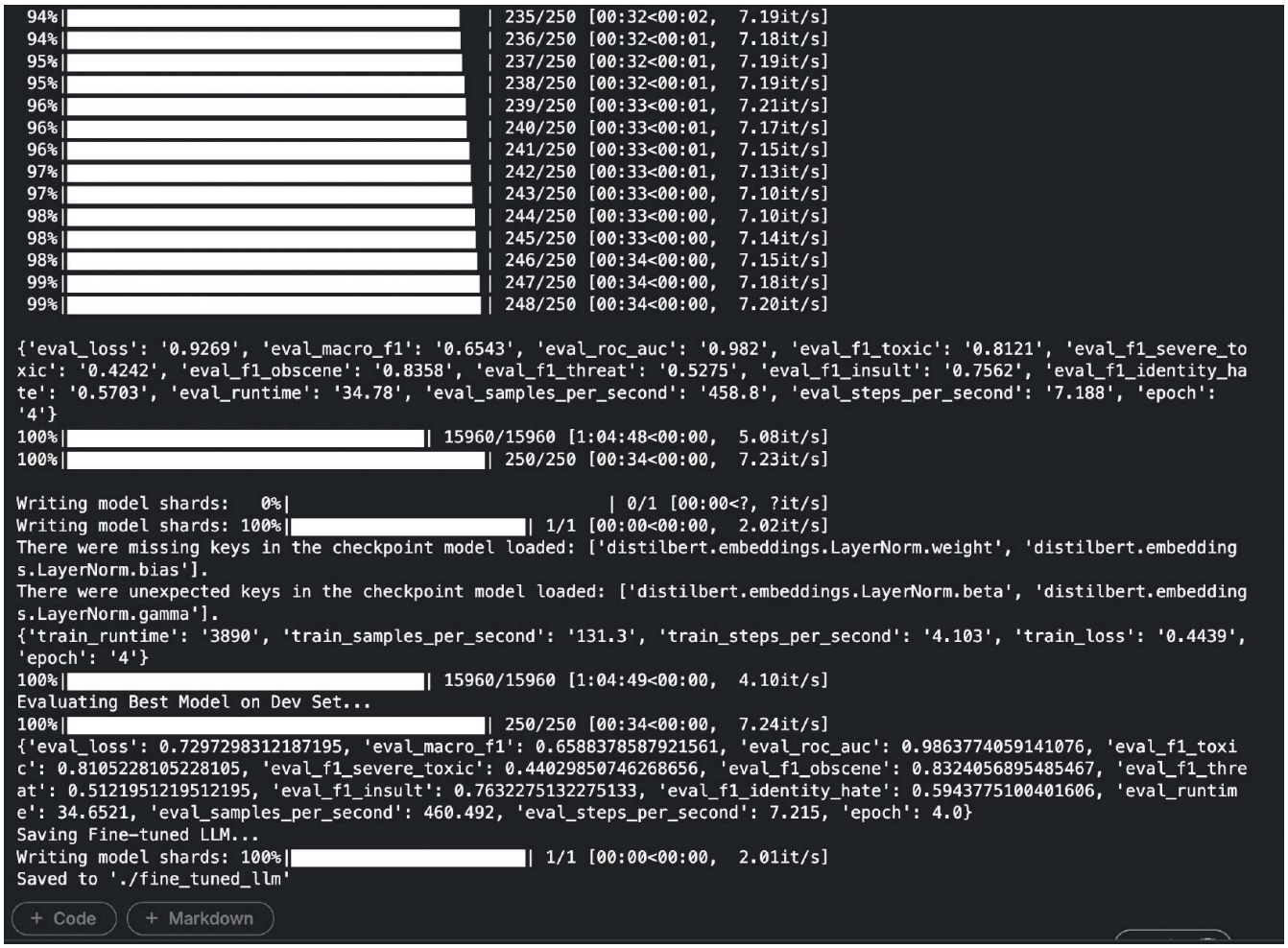

Notebook output — training loop running (epoch 0 → 0.48)



Notebook output — epoch 2 evaluation, training continues



Notebook output — epoch 4 evaluation, training complete, model saved

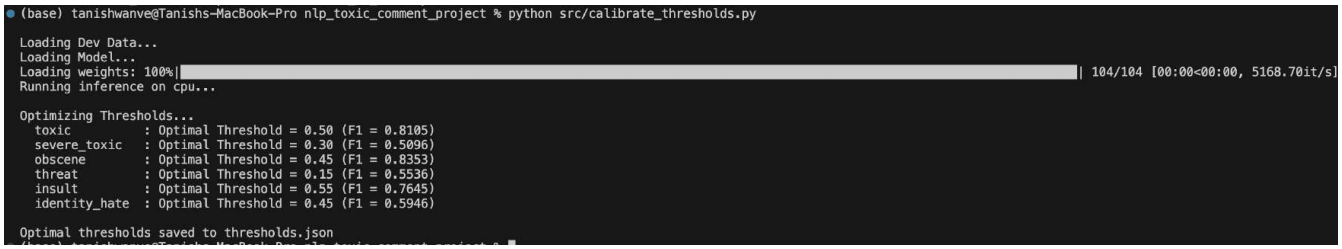
Result

The pretrained distilbert-base-uncased checkpoint loaded with the expected MISSING/UNEXPECTED key warnings (normal, since the classification head isn't part of the pretrained weights). Training ran for 4 epochs (train_runtime ≈ 3,890s), with the training loss falling from ≈14 at the very first step down to a final train_loss of 0.4439. On the dev set, the best checkpoint reached a macro F1 of 0.6588 and ROC-AUC of 0.9864, with per-label F1 scores of: toxic 0.81, severe_toxic 0.44, obscene 0.83, threat 0.51, insult 0.76, and identity_hate 0.59. The fine-tuned model was saved to `./fine_tuned_llm`.

Step 7: Threshold Calibration

src/calibrate_thresholds.py loads the fine-tuned model, runs it on the dev set, and sweeps a Precision-Recall curve per label to find the decision threshold that maximizes F1 for each of the six labels individually, instead of using one flat 0.5 cutoff for everything.

Command: `python src/calibrate_thresholds.py`



Terminal output — python src/calibrate_thresholds.py

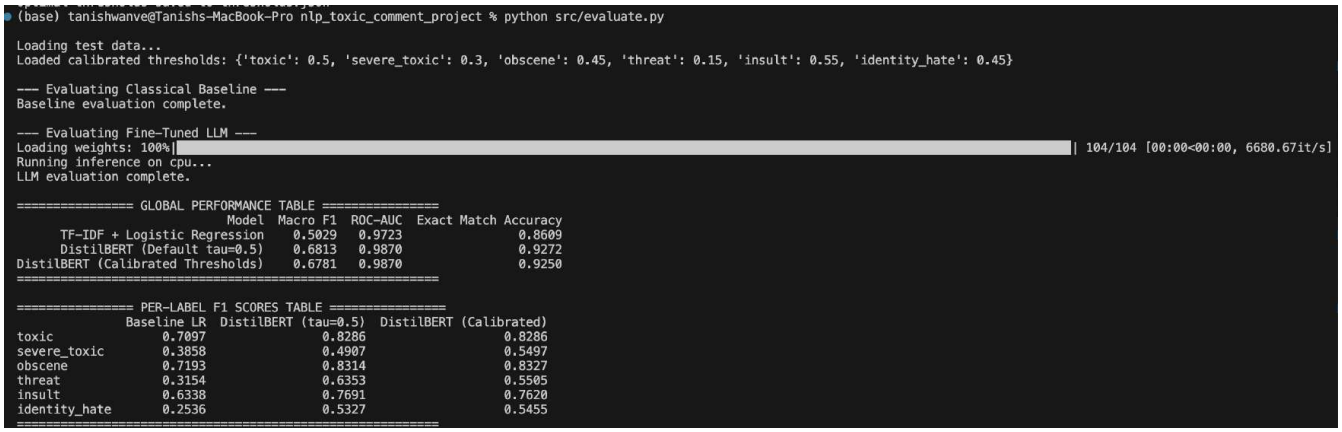
Result

The optimal per-label thresholds found were: toxic 0.50, severe_toxic 0.30, obscene 0.45, threat 0.15, insult 0.55, and identity_hate 0.45. The rarer labels (threat, severe_toxic, identity_hate) ended up with noticeably lower thresholds than 0.5, which makes sense given how few positive examples they have in training. These values were saved to `thresholds.json` for use during evaluation and inference.

Step 8: Final Model Evaluation

src/evaluate.py is the final, fair comparison: it scores both the classical TF-IDF + Logistic Regression baseline and the fine-tuned DistilBERT model on the held-out test set (data neither model has seen before), at both the default 0.5 threshold and the calibrated per-label thresholds from Step 7.

Command: `python src/evaluate.py`



Terminal output — python src/evaluate.py

Result

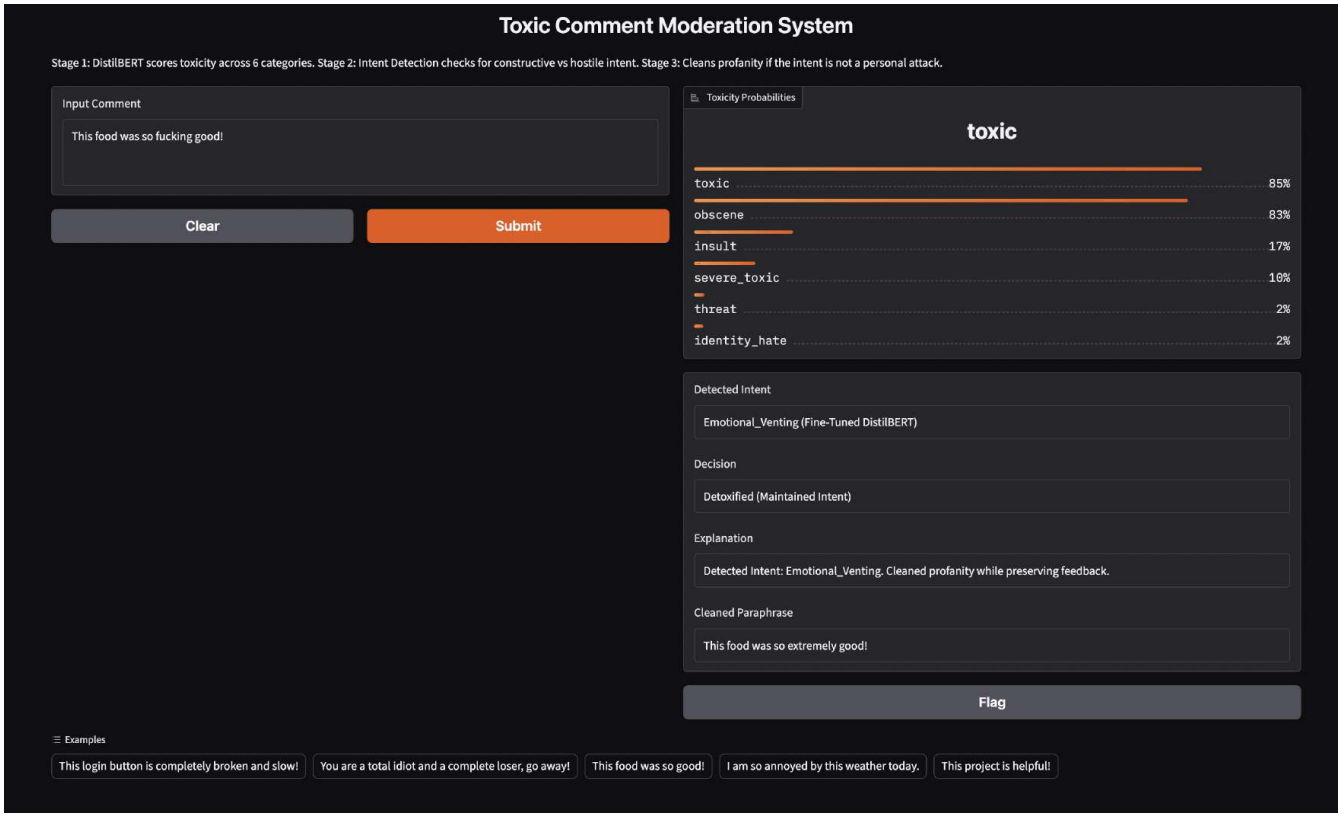
DistilBERT clearly beats the classical baseline: macro F1 rises from 0.5029 (baseline) to 0.6813 (DistilBERT, default threshold), and ROC-AUC improves from 0.9723 to 0.9870. Moving to calibrated per-label thresholds (0.6781 macro F1) costs a negligible amount of overall macro F1 in exchange for large gains on the rarest classes — identity_hate F1 rises from 0.2536 (baseline) to 0.5455 (calibrated DistilBERT), and threat rises from 0.3154 to 0.5505.

Step 9: Live System Demo (Web App)

app.py ties all three pipeline stages together into an interactive Gradio app: Stage 1 (DistilBERT) scores each of the six toxicity labels, Stage 2 checks whether the comment is a hostile personal attack versus something more benign like venting or critique, and Stage 3 either blocks the comment outright (for direct personal attacks) or rewrites it into a cleaned, non-toxic paraphrase (for everything else).

Command: `python app.py` → opened at `http://127.0.0.1:7860`

Example 1: "This food was so fucking good!"

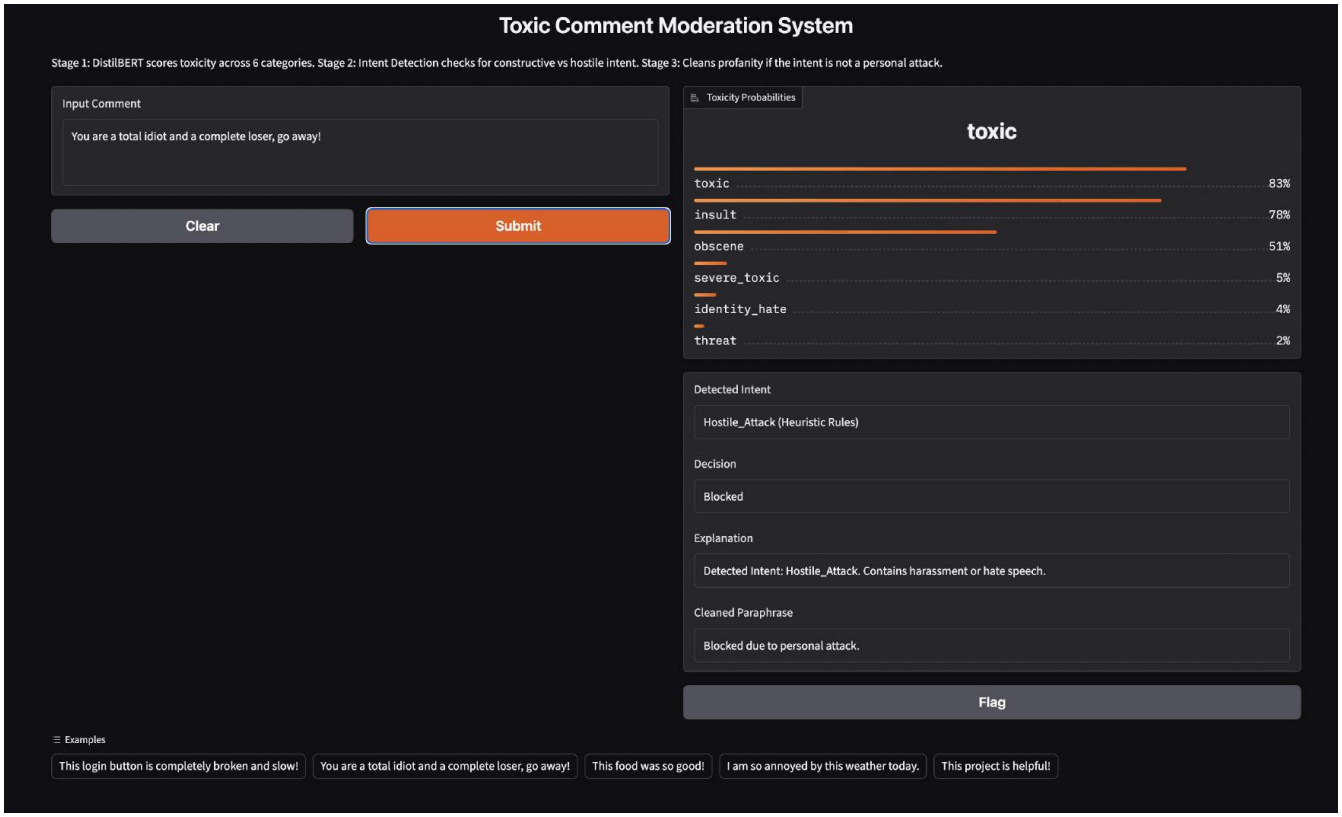


Web UI — venting comment, detoxified rather than blocked

Result

Scored toxic 85% and obscene 83%, but Stage 2 classified the intent as Emotional_Venting rather than an attack, so Stage 3 detoxified it instead of blocking it, producing: "This food was so extremely good!" — the positive sentiment is preserved, only the profanity is removed.

Example 2: “You are a total idiot and a complete loser, go away!”

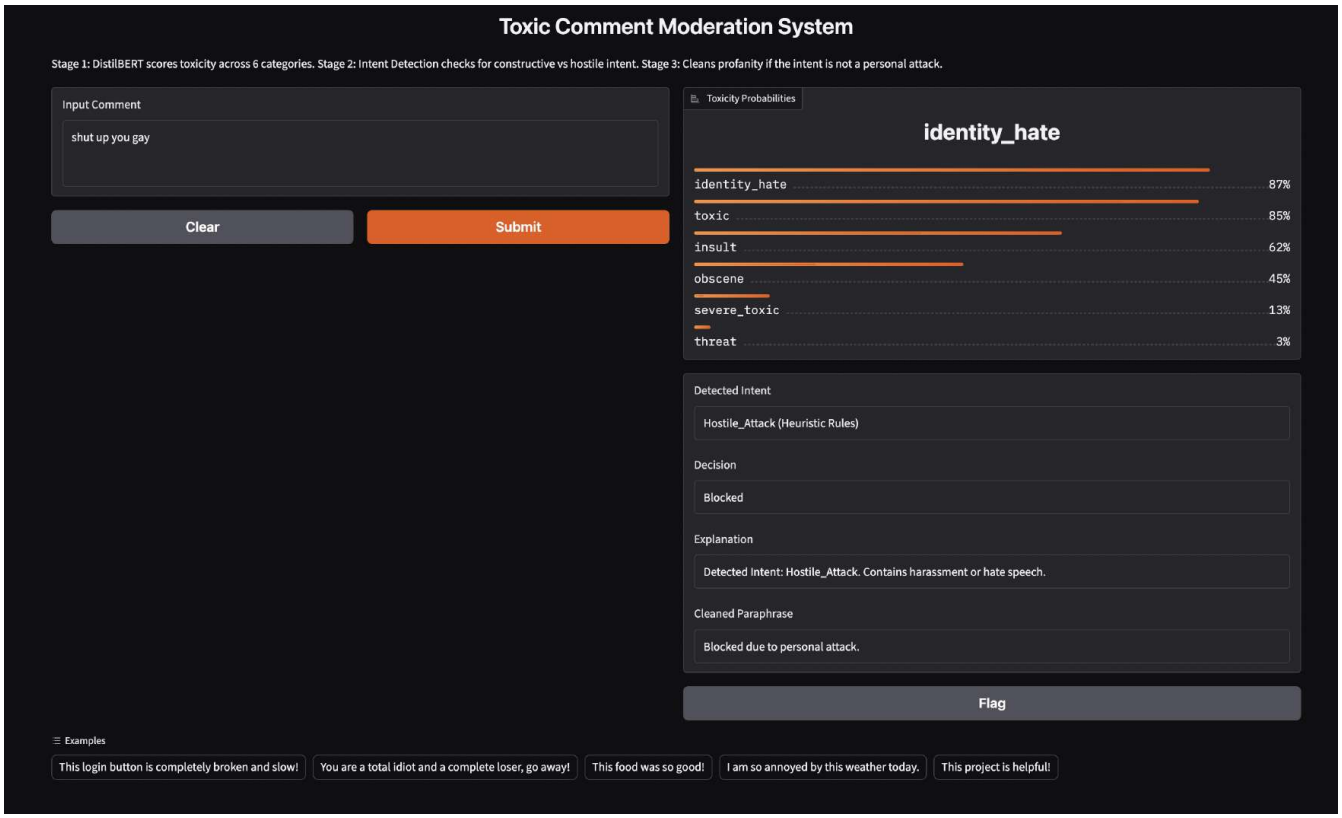


Web UI — direct personal attack, blocked

Result

Scored toxic 83%, insult 78%, obscene 51%. Stage 2's heuristic rules flagged this as a Hostile_Attack, so Stage 3 blocked the comment outright instead of trying to clean it.

Example 3: “shut up you gay”

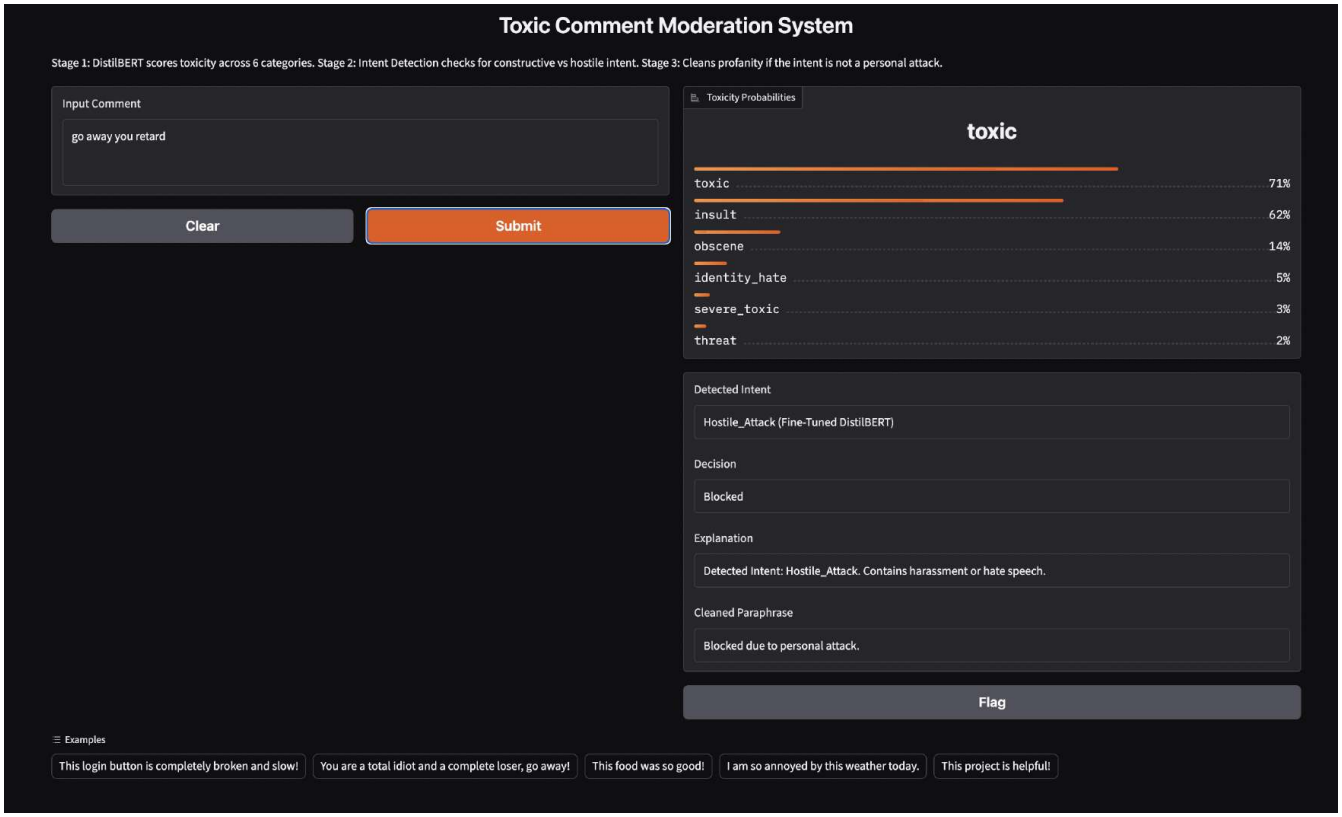


Web UI — identity-based hate speech, blocked

Result

identity_hate was the top-scoring label at 87% (toxic 85%, insult 62%). Again classified as a Hostile_Attack and blocked, showing the moderation layer also catches identity-based hate speech, not just generic insults.

Example 4: “go away you retard”



Web UI — ableist slur, blocked by the fine-tuned intent classifier

Result

Scored toxic 71%, insult 62%. This time it was Stage 2's fine-tuned DistilBERT intent classifier (rather than the heuristic rules) that flagged Hostile_Attack, and the comment was blocked.

Overall takeaway

Across all four examples the moderation layer behaves as intended: comments that vent frustration or express strong sentiment without targeting a person are detoxified and handed back to the user, while direct personal attacks — whether plain insults or identity-based slurs — are blocked outright, regardless of whether the heuristic rules or the fine-tuned classifier is the one that catches them.